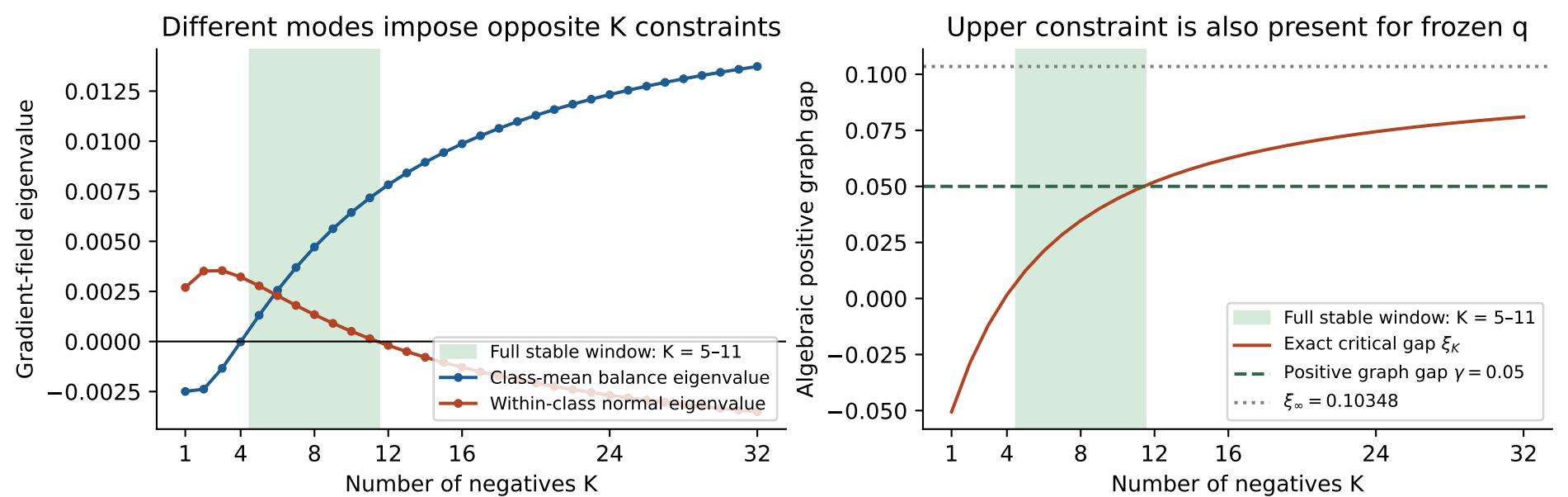
Different modes impose opposite K constraints
Gradient-field eigenvalue
0.0125
0.0100
0.0075
0.0050
0.0025
0.0000
−0.0025
1
4
8
12
16
24
32
Number of negatives K
Full stable window: K = 5–11
Class-mean balance eigenvalue
Within-class normal eigenvalue
Upper constraint is also present for frozen q
Algebraic positive graph gap
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
1
4
8
12
16
24
32
Number of negatives K
Full stable window: K = 5–11
Exact critical gap ξ_K
Positive graph gap $\gamma = 0.05$
$\xi_\infty = 0.10348$